

Unit 8, Lesson 4: Area of Composite Figures

Benchmark

MA.6.GR.2.2 Solve mathematical and real-world problems involving the area of quadrilaterals and composite figures by decomposing them into triangles or rectangles.

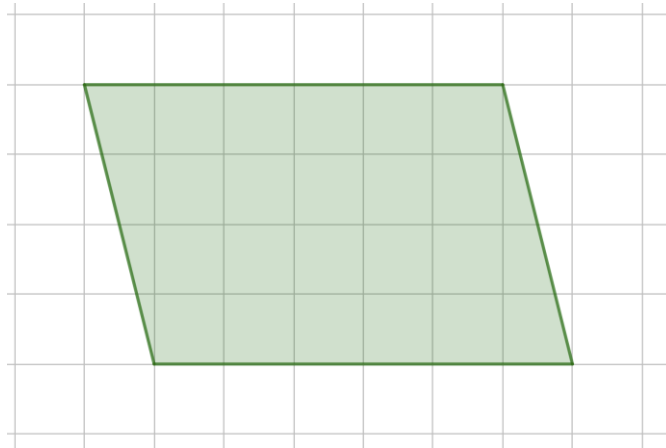
Learning Target

- ☐ I can find the area of composite figures.

8.4.1 Warm-Up: Bell Work

In the figure, each square of the grid paper is 1 centimeter (cm) by 1 cm.

1. Find the area of the shaded region. Include units.

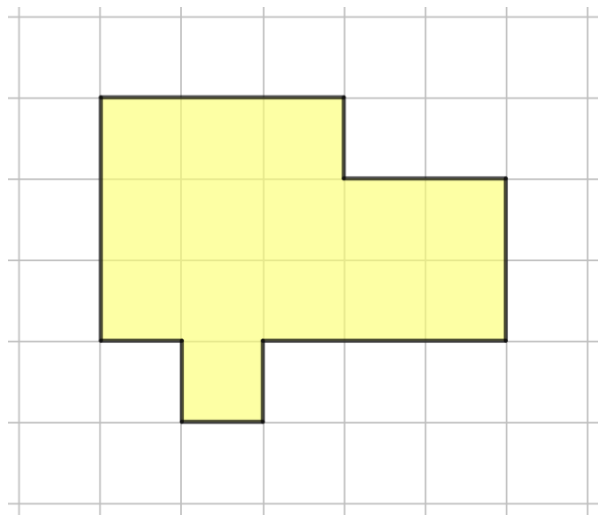


8.4.2 Guided Instruction: Area of Composite Figures



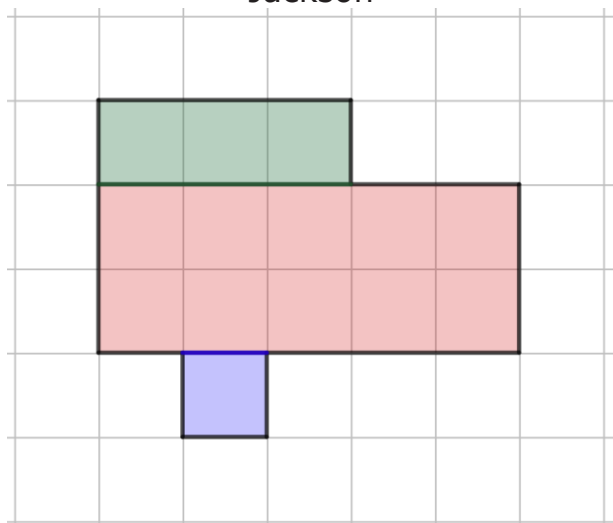
A **composite figure** is a two- or three- dimensional figure that can be decomposed into smaller figures.

Jackson and Danna were instructed to find the area of the composite figure below. The length of each square is 1 centimeter (cm).

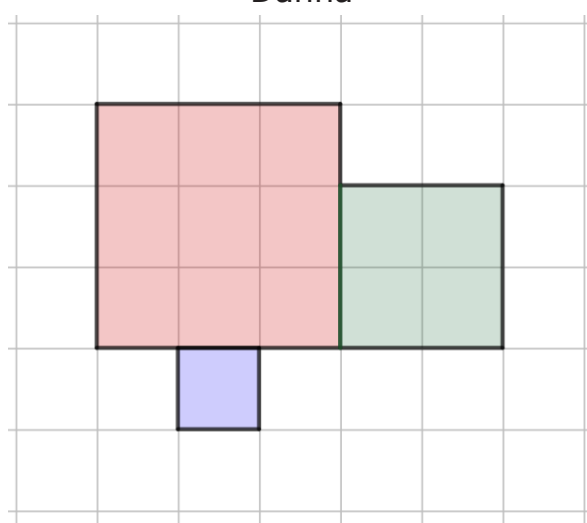


Both Jackson and Danna decomposed the figure, but their teacher noticed they each decomposed it differently.

Jackson



Danna



1. Complete the table about each student’s work. Include units.

Student	Area of Red Rectangle	Area of Blue Rectangle	Area of Green Rectangle
Jackson			
Danna			

2. To find the area of the composite figure, find the sum of the areas of the rectangles for each student. Include units.

	Area of the Composite Figure
Jackson	
Danna	

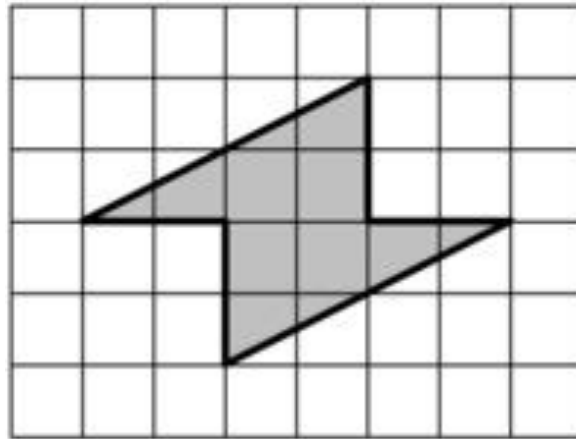
3. Explain why both students got the same area of the composite shape.



The **area of a composite figure** is the sum of the areas of the figure’s decomposed parts.

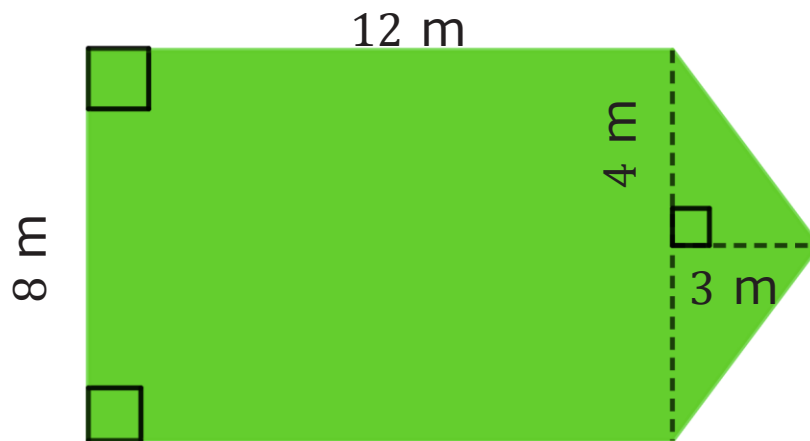
8.4.3 Guided Practice: Area of Composite Figures

1. A composite shape is shown. The length of each square is 1 inch (in).



- Draw lines to decompose the figure into rectangles and/or triangles.
- Find the area of each rectangle and/or triangle. Include units.
- Find the sum of the areas of the rectangles and/or triangles to find the area of the composite figure. Include units.
- Find a classmate who decomposed the shape differently than you did. Describe their process.

2. Thomas has decided to put grass sod in his backyard. A drawing of his backyard is shown.

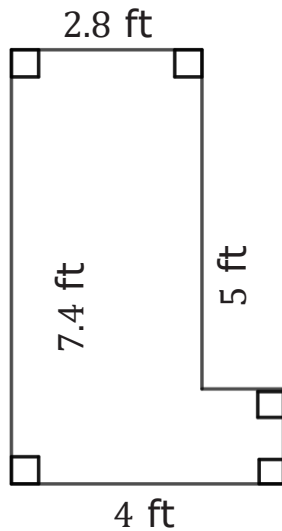


How much sod does Thomas need to buy?

8.4.4 Your Turn!

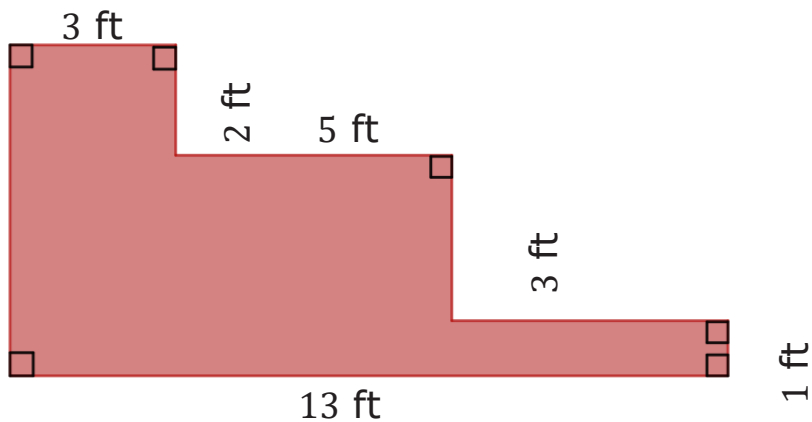
For each composite figure, find the area, in square feet (ft^2). Show your work.

1.



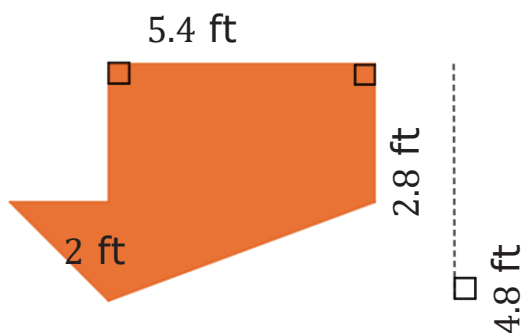
Area = _____

2.



Area = _____

3.



Area = _____

8.4.5 Wrap-Up: Reflection

1. Plot a point in one of the quadrants below to indicate how you feel about today's learning target: "I can find the area of composite figures."

